A Closure Geometry on the 600-Cell and Its Arithmetic Shadow

The icosian object, its L-function, and a certified arithmetic boundary

(VFD programme — working draft)

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Abstract

We study a single geometric object—the icosian closure structure on the 600-cell— and ask what arithmetic it carries. The object is rigorous geometry: the 120 unit icosians, the maximal order $\mathcal{I}$ they generate over $K = \mathbb{Q}(\sqrt{5})$, and its E_8 lattice. Its *arithmetic shadow* is the L-function of its theta series, which (by prior work of the programme) is identified as $\zeta_K(s)\zeta_K(s-1)$ by the classical Siegel–Weil / maximal-order Eisenstein framework and checked by complete short-vector enumeration of the local representation data. (The theorem carries the identity; the computation verifies the implementation, not an infinite Euler product by itself.) We then map the *boundary* of this shadow: using a reproducible verification gate, we certify that the object's L-function contains the Riemann ζ (the unshifted factor $\zeta(s)$) only as one factor among four, and that the proposed strong correspondence "object $\leftrightarrow \zeta$" is rejected. Finally we record, as clearly-marked *interpretation*, the VFD physical reading of the same object as a compactified scale–phase system whose closure condition is a positivity witness. **No claim is made about the Riemann Hypothesis**, and no cosmological claim is made or implied. The mathematics is geometry-first; the physics is the frame, not the proof.

1 The object

This programme has been geometry-first throughout: the primitive is a shape, and its arithmetic is discovered as a consequence. The shape here is the *icosian closure object* on the 600-cell—the regular 4-polytope whose 120 vertices are the unit icosians, i.e. the binary icosahedral group $2I$. These generate the icosian ring $\mathcal{I}$, a maximal order in the definite quaternion algebra over $K = \mathbb{Q}(\sqrt{5})$ ramified at the two infinite places; its underlying $\mathbb{Z}$-lattice is a copy of E_8 [1].

Within the VFD programme this object is read physically as a compactified scale–phase–witness system (§5); that reading motivates the study but is *not* used in any of the mathematics of §§2–4. The companion documents `from-a-point-to-a-universe` (intuition) and `vfd-rh-reformulation` (a falsifiable reformulation) carry the physical narrative in full; this paper keeps it walled to §5.

What is claimed. The geometry of §2; the arithmetic shadow of §3 (the exact icosian identity $L(\vartheta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)$, recalled from [3]); and the certified boundary of §4 (the object's L-function contains ζ only as a factor; the strong "object $\leftrightarrow \zeta$" bridge candidate is rejected by an explicit gate).

What is not claimed. Nothing about the location of the zeros of ζ (the Riemann Hypothesis); no positivity, self-adjointness, operator, or arithmetic-surface *result*; no connection between any physical or cosmological model and RH. The boundary of §4 concerns a specific *construction*; it implies nothing about RH.

2 The geometry

Let $\varphi = \frac{1+\sqrt{5}}{2}$, $\mathcal{O}_K = \mathbb{Z}[\varphi]$. The Dedekind zeta of $K = \mathbb{Q}(\sqrt{5})$ factors as

$$\zeta_K(s) = \zeta(s)\, L(s, \chi_5), \tag{1}$$

with χ_5 the even real quadratic character mod 5 [2]. The ring $\mathcal{I}$ is the $\mathbb{Z}[\varphi]$-order on the 120 unit icosians; its 24-element rational sub-structure is the group of Hurwitz units (the 24-cell / $2T$, over $\mathbb{Q}$), and the remaining 96 vertices form the φ-shell. The coordinate Galois involution $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ fixes the 24-cell pointwise and permutes the φ-shell. Throughout, "the object" (or "substrate") refers to $\mathcal{I}$ with this sub-structure.

Lemma 1 (class number one)**.** *The icosian maximal order $\mathcal{I}$ in the definite quaternion algebra over $K = \mathbb{Q}(\sqrt{5})$ ramified exactly at the two infinite places has a single (left- or right-) ideal class.*

Reference. $\mathcal{I}$ is the maximal-order case with invariants $(n, d_F, D, N) = (2, 5, 1, 1)$ in the Kirschmer–Voight enumeration of class-number-one definite quaternion orders over totally real fields [4, Table 8.2] (see also Vignéras [5]). Left and right class numbers agree here by conjugation. □

3 The arithmetic shadow

The arithmetic carried by the object is the L-function of its theta series. It is *exactly* the rank-two product $\zeta_K(s)\zeta_K(s-1)$, verified in [3] by complete short-vector enumeration in the maximal icosian order.

Proposition 2 (recalled from [3])**.** *Let $\vartheta_{\mathcal{I}}$ be the Hilbert theta series over K of the norm form on the maximal icosian order $\mathcal{I}$ (weight two). By the Siegel–Weil formula [6, 7] and Lemma 1 (single class), $\vartheta_{\mathcal{I}}$ is the corresponding Eisenstein series with no cuspidal contribution, and*

$$L(\vartheta_{\mathcal{I}}, s) = \zeta_K(s)\, \zeta_K(s-1) \qquad \textit{exactly.}$$

The maximal-order representation numbers $r(\pi) = 120(1 + N_Q(\pi))$ at every prime (120 = number of unit icosians), and $r((2)^k)/120 = \sigma_1((2)^k) = 1, 5, 21, 85$ at the inert prime (2), give the standard Eisenstein local factor everywhere; there is no local correction.

Remark 3 (on the earlier "local-2 factor")**.** An earlier draft of [3] proposed a correction $C_2(s) \neq 1$ at the prime 2, from a count $r(2) = 24$. That value is the count in the *Lipschitz suborder* $\mathbb{Z}[\varphi]^4$ (8 units), a bounded-range enumeration proxy that is non-maximal at 2; for the maximal icosian order $r(2) = 600 = 120 \cdot 5$ and the local factor is standard, so $C_2 \equiv 1$. The correction has been withdrawn (it was also dimensionally ill-formed: an inert prime of norm 4 admits a local factor only in 4^{-s}). The identity above is exact.

The two Galois components of the object are tested against the two factors of (1):

Numerical Check 4. The rational 24-cell sub-object (Hurwitz, over $\mathbb{Q}$) is the $K = \mathbb{Q}$ analogue: its theta series has on-line zeros matching those of $\zeta(s)\zeta(s-1)$ (tested to finite precision; the same Siegel–Weil mechanism applies, the order being class-number one over $\mathbb{Q}$). This is a recorded consistency check, not a theorem-grade derivation here.

Proposition 5. *On the line* $\mathrm{Re}(s) = \frac{1}{2}$*, the zeros of $\zeta(s)\zeta(s-1)$ are exactly those nontrivial zeros of ζ that lie on that line; $\zeta(s-1)$ contributes none.*

Proof. A nontrivial zero of $\zeta(s-1)$ sits at $s = 1+\rho$, $\mathrm{Re}(\rho) \in (0,1)$, hence $\mathrm{Re}(s) \in (1,2)$; trivial zeros of $\zeta(s-1)$ are at $s = 1-2n$. Neither meets $\mathrm{Re}(s) = \frac{1}{2}$. The statement concerns *which* zeros lie on the line and asserts nothing about whether all nontrivial zeros of ζ do. □

Numerical Check 6. The 96-vertex φ-shell (Galois-sign component) has tested on-line zeros matching those of $L(s, \chi_5)$, with no observed coincidence with a Riemann zero in the tested range. This too is a recorded consistency check, not a geometric derivation here.

4 The boundary

The achievement of this section is *not* that the object contains ζ—many constructions do, and that alone is not special. It is the pair: the object carries a precise Dedekind/Eisenstein arithmetic shadow, *and* the stronger "object $= \zeta$" identity provably fails. The object's L-function (Proposition 2) has degree 4 and, by (1), equals $\zeta(s)L(s,\chi_5)\zeta(s-1)L(s-1,\chi_5)$, with no local-2 correction; the unshifted $\zeta(s)$ is one factor among four (the shifted $\zeta(s-1)$ is also present but is irrelevant to the critical-line count, by Proposition 5). Whether the object *isolates* ζ is the real question; it does not, and we certify this.

The gate. A proposed correspondence supplies a parameter-free procedure producing the on-line zeros of a claimed L-function *without taking the target as input.* The gate checks **fingerprint** (nearest-neighbour spacing: fraction of normalized gaps below 0.3), **pointwise** (first eight zeros, tolerance 0.2), **density** (on-line zeros per unit height), **rigidity** (number variance), structural **degree/pole**, and—auxiliary only—**no-hardcoding / provenance** (declared-procedure checks that catch a *declared* fitted map, relying on honest declaration, not program analysis). **The decisive rejections below are structural** (degree, density, pointwise); provenance is a guardrail, not the proof. A hashed certificate is emitted either way; the engine is a certifier, not a search. The geometry→arithmetic enumeration itself is the separate computation of [3], not re-performed by the gate.

Gate Result 7 (the ζ-boundary)**.** The strong candidate "object $\leftrightarrow \zeta$" is **rejected**. Against ζ (degree 1) the object's degree-4 on-line content fails: *degree* (4 vs 1); *fingerprint* (0.077 for the superposition vs 0.000 for a single ζ-type spectrum); *pointwise* (maximum displacement over the first eight zeros ≈ 22.3; first-zero displacement ≈ 7.5); and *density* (Result 8).

Gate Result 8 (counting route)**.** A route sharing only the density diagnostic with Result 7: in the computed on-line lists, the object's combined zero count exceeds ζ's—the count ratios are $\approx 4.7, 3.7, 3.1$ in the windows up to heights $30, 40, 50$. (A leading-order doubled density is the heuristic expectation for the two unshifted degree-one factors visible on this line; we do not certify the asymptotic, and present this as finite-window evidence, not a theorem.) A list denser than ζ's on these windows is not ζ's.

Controls.

Gate Result 9 (fitted-map control)**.** A candidate whose provenance *declares* a least-squares fit to the zeros is rejected on **provenance** alone, independent of fit quality—reproducing, mechanically, the fitted-map trap of a prior internal audit (the "$W5$" audit; see `CIRCLE_TEST.md`).

Gate Result 10 (class-vs-individual control)**.** A single seeded, window-rescaled GUE operator (random Hermitian, no arithmetic; produced by `grow_atlas.py`) *passes* the **fingerprint** check—right universality class—but *fails* **pointwise** (wrong individual spectrum); in the saved atlas it also fails density and rigidity, though those depend on a window-scaling convention and are not advanced as intrinsic discriminators. The load-bearing rejection is pointwise: GUE-class membership is insufficient to be ζ. This is one control instance, not a theorem about generic GUE operators.

candidate	verdict	load-bearing rejection
24-cell $\to \zeta(s)\zeta(s-1)$ (on-line)	CONSISTENT	—
icosian $\to \zeta_K(s)\zeta_K(s-1)$ (exact)	VERIFIED	—
φ-shell $\to L(s, \chi_5)$ (on-line)	CONSISTENT	—
object $\to \zeta$ (strong bridge)	REJECTED	degree, fingerprint, pointwise, density
fitted map $\to \zeta$	REJECTED	provenance
seeded GUE $\to \zeta$	REJECTED	pointwise (class $\neq$ individual)

("Consistent" = passes the gate's declared-consistency checks against the named target. The icosian row rests on [3]; the 24-cell and φ-shell rows are finite consistency checks, not theorem-grade derivations.)

The net statement is a *boundary* on the object's arithmetic shadow: it carries $\zeta_K(s)\zeta_K(s-1)$ exactly, in which the unshifted $\zeta(s)$ is one factor, and it does not—certifiably, for this candidate and this gate—isolate ζ.

5 The physical reading (interpretation)

This section is interpretation; nothing above depends on it, and it proves nothing. VFD reads the same object as a single compactified scale–phase system with a positivity witness,

$$\mathcal{V}_1 = \big([-\infty, +\infty] \times S^1,\ \pi,\ Q\big), \qquad \pi(t, \theta) = e^{t/2} e^{i\theta},$$

in which the point ($t \to -\infty$), the unit circle ($t = 0$), and infinity ($t \to +\infty$) are three boundary cuts of one object, and the critical line $\mathrm{Re}(s) = \frac{1}{2}$ is read as the balanced equator. The witness Q is the object's closure law, schematically $Q = H - R$ (archimedean/boundary capacity minus prime residue), and the VFD reading of RH is the closure condition $Q[\Phi] \geq 0$.

Two honest points fix the status of this reading. First, with Q the Weil explicit-formula functional, "$Q \geq 0 \iff$ RH" is the *Weil criterion*—a known theorem, restated in scale–phase coordinates, not new. Second, the only non-trivial step would be to *construct* a witness operator A with $Q(f) = \langle f, Af \rangle$ geometrically from $\mathcal{V}_1$ and show $A \geq 0$; that positivity *is* RH and is not claimed here. The reading is, however, *falsifiable*: one searches for f with $Q(f) < 0$; a single such f would refute it. The companion `vfd-rh-reformulation` develops this with the construction target and the test stated precisely. In the slogan of the picture: *geometry gives the object; arithmetic tests its closure.*

6 Scope

We claim the geometry of §2, the recalled arithmetic shadow of §3 (with no local-2 correction; including the inert-prime-2 check), and the certified boundary of §4. We make no claim about the Riemann Hypothesis: the unshifted $\zeta(s)$ enters only as one factor of the object's L-function, and the negative is a statement about a construction, not about the zeros of ζ. The physical reading of §5 is interpretation. The gate, certificates, and atlas accompany this paper in `vfd_math_engine/` (the seeded GUE control row is produced by `grow_atlas.py`). No proof certificate is defined or issued.

References

[1] J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, Springer.

[2] J. Neukirch, *Algebraic Number Theory*, Springer.

[3] VFD programme, *The Icosian Triad on* V_{600} (`icosian-triad-v600` release, v1.1); verifies $L(\vartheta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)$ exactly by complete short-vector enumeration in the maximal icosian order ($r(\pi) = 120(1 + N_Q(\pi))$ at every prime; $C_2 = 1$).

[4] M. Kirschmer, J. Voight, *Algorithmic enumeration of ideal classes for quaternion orders*, SIAM J. Comput. **39** (2010), Table 8.2.

[5] M.-F. Vignéras, *Arithmétique des algèbres de quaternions*, Lecture Notes in Math. **800**, Springer (1980).

[6] C. L. Siegel, *Über die analytische Theorie der quadratischen Formen*, Ann. of Math. (1935).

[7] A. Weil, *Sur la formule de Siegel...*, Acta Math. (1965).